Unit: Unit 3: Functions **Name:** ________________

Topic: Inverse Relations and Functions **Date:** ___________

Advanced (Carolina Reaper)

Q1. Which of the following functions has an inverse?
(A) $f(x) = x^2$
(B) $f(x) = x^3$
(C) $f(x) = |x|$
(D) $f(x) = 2x$
(E) $f(x) = x - 1$

Q2. Find the inverse of $f(x) = 3x - 2$. Let $f^{-1}(x)$ be the inverse function.
(A) $f^{-1}(x) = \frac{x+2}{3}$
(B) $f^{-1}(x) = \frac{x-2}{3}$
(C) $f^{-1}(x) = \frac{x+2}{2}$
(D) $f^{-1}(x) = \frac{x-2}{2}$
(E) $f^{-1}(x) = \frac{x+3}{2}$

Q3. If $f(x) = 2x$ and $g(x) = \frac{x}{2}$, which statement is true?
(A) $f(g(x)) = x$ and $g(f(x)) = x$
(B) $f(g(x)) = x$ but $g(f(x)) \neq x$
(C) $f(g(x)) \neq x$ but $g(f(x)) = x$
(D) $f(g(x)) \neq x$ and $g(f(x)) \neq x$
(E) $f(g(x)) = 2x$ and $g(f(x)) = 2x$

Q4. Tom has been saving money for a new bike and has $120inhissavingsaccount.Hewantstobuyabikethatcosts180$. If Tom's savings account earns interest at a rate of $f(x) = 0.05x$, how much will he have after 1 year?
(A) $f(120) = 6$
(B) $f(120) = 12$
(C) $f(120) = 18$
(D) $f(120) = 24$
(E) $f(120) = 30$

Q5. Which of the following statements about inverse functions is true?
(A) The domain of the inverse function is the same as the range of the original function.
(B) The range of the inverse function is the same as the domain of the original function.
(C) The domain and range of the inverse function are the same as the domain and range of the original function.
(D) The inverse function is always a linear function.
(E) The inverse function is never a linear function.

Q6. A water tank can be filled at a rate of $f(x) = 2x$ gallons per minute. If $g(x)$ is the inverse function of $f(x)$, what does $g(x)$ represent?
(A) The amount of time it takes to fill the tank with x gallons of water.
(B) The amount of water in the tank after x minutes.
(C) The rate at which the tank is filled.
(D) The capacity of the tank.
(E) The amount of water that can be filled in x minutes.

Q7. A company has a function $f(x)$ that represents the cost of producing x units of a product. If the inverse function is $f^{-1}(x)$, what does $f^{-1}(x)$ represent?
(A) The number of units produced at a cost of x dollars.
(B) The cost of producing x units.
(C) The profit made from selling x units.
(D) The number of units sold at a price of x dollars.
(E) The revenue generated from selling x units.

Q8. Which of the following is a real-world application of inverse functions?
(A) Calculating the area of a circle.
(B) Finding the inverse of a linear function.
(C) Determining the cost of producing x units of a product.
(D) Scheduling appointments.
(E) Decoding secret messages.

Open Ended Questions (FRQ)

Q9. A person is planning a road trip from City A to City B. The distance between the two cities is 240 miles, and the person wants to drive at an average speed of 40 miles per hour. If $f(x)$ represents the time it takes to drive x miles, find the inverse function $f^{-1}(x)$ and interpret its meaning in the context of the problem. Show all work and explain your answer.

Q10. A company has a function $f(x) = 2x+1$ that represents the amount of money (in dollars) a person spends on groceries per week. Find the inverse function $f^{-1}(x)$ and use it to determine how much a person spends if they buy $12worthofgroceries.Showallworkandex$

Chef's Tips

- Tip 1: Encourage students to check their work by plugging their answers back into the original equation.
- Tip 2: Remind students that the inverse of a function is not always a function itself.
- Tip 3: Emphasize the importance of understanding the context and interpreting the results in real-world applications.